

Practical-11

AIM: Develop an Android application that uses Services to perform background tasks.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical11_layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical11">
<Button
    android:id="@+id/Practical11_start_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Start the service"

app:layout_constraintBottom_toTopOf="@id/Practical11_stop_btn"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />
<Button
    android:id="@+id/Practical11_stop_btn"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Stop the service"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Practical11_start_btn" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Service.java

```
package com.mk.madpractical;

import android.app.Service;
import android.content.Intent;
import android.os.Binder;
import android.os.Handler;
import android.os.Looper;
```

```
import android.os.IBinder;
import android.widget.Toast;

public class Practical11Service extends Service {

    private Thread thread;
    private boolean isServiceRunning;

    public Practical11Service() {
    }

    @Override
    public IBinder onBind(Intent intent) {
        return new Binder() {
            Practical11Service get_service() {
                return Practical11Service.this;
            }
        };
    }

    @Override
    public void onCreate() {
        super.onCreate();
        isServiceRunning = true;

        // Create and start a new thread
        thread = new Thread(new Runnable() {
            @Override
            public void run() {
                while (isServiceRunning) {
                    new
                    Handler(Looper.getMainLooper()).post(new Runnable() {
                        @Override
                        public void run() {
                            Toast.makeText(Practical11Service.this,
                                "Service is running", Toast.LENGTH_SHORT).show();
                        }
                    });
                }
            }
        });

        try {
            Thread.sleep(5000);
        } catch (InterruptedException e) {
            e.printStackTrace();
        }
    }

    @Override
    public int onStartCommand(Intent intent, int flags, int
```

```

startId) {
    if (!thread.isAlive()) {
        thread.start(); // Start the thread
    }
    return START_STICKY; // Keep the service running
}

@Override
public void onDestroy() {
    super.onDestroy();
    isServiceRunning = false; // Stop the loop
    if (thread != null && thread.isAlive()) {
        thread.interrupt(); // Safely stop the thread
    }
}
}

```

Practical11.java

```

package com.mk.madpractical;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical11 extends AppCompatActivity {
    Button start,stop;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

```

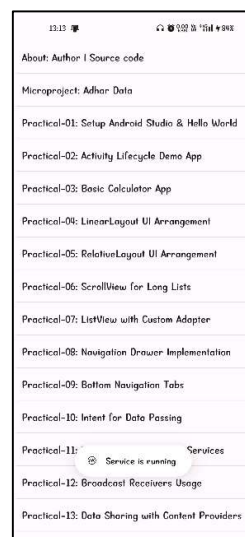
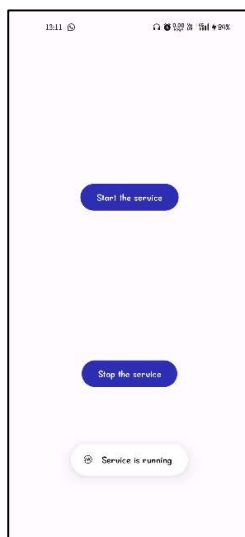
```

        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical11);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical11_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        start = findViewById(R.id.Practical11_start_btn);
        stop = findViewById(R.id.Practical11_stop_btn);
        Intent service = new
        Intent(this,Practical11Service.class);
        View.OnClickListener listener = new
        View.OnClickListener() {
            @Override
            public void onClick(View v) {

                try {
                    if (v.equals(start)){
                        startService(service);
                    }else {
                        stopService(service);
                    }
                } catch (Exception e) {
                    throw new RuntimeException(e);
                }
            }
        };
        start.setOnClickListener(listener);
        stop.setOnClickListener(listener);
    }
}

```



Practical-12

AIM: Develop an Android application that uses Broadcast Receivers to receive and handle system-level broadcasts.

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical12_ConstraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical12">

    <TextView
        android:id="@+id/Practical12_textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="plug the power cable or take it off to
see the broadcast"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.content.IntentFilter;
import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
```

```
public class Practical12 extends AppCompatActivity {
    Practical12_broadcast broadcast;
    IntentFilter intentFilter;
    TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical12);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Practical12_ConstraintLayout), (v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
            return insets;
        });

        textView = findViewById(R.id.Practical12_textview);

        broadcast = new Practical12_broadcast();
        intentFilter = new IntentFilter();

        intentFilter.addAction("android.intent.action.ACTION_POWER_CONNECTED");

        intentFilter.addAction("android.intent.action.ACTION_POWER_DISCONNECTED");
        registerReceiver(broadcast, intentFilter,
Context.RECEIVER_NOT_EXPORTED);

    }

    @Override
    protected void onDestroy() {
        unregisterReceiver(broadcast);
        super.onDestroy();
    }

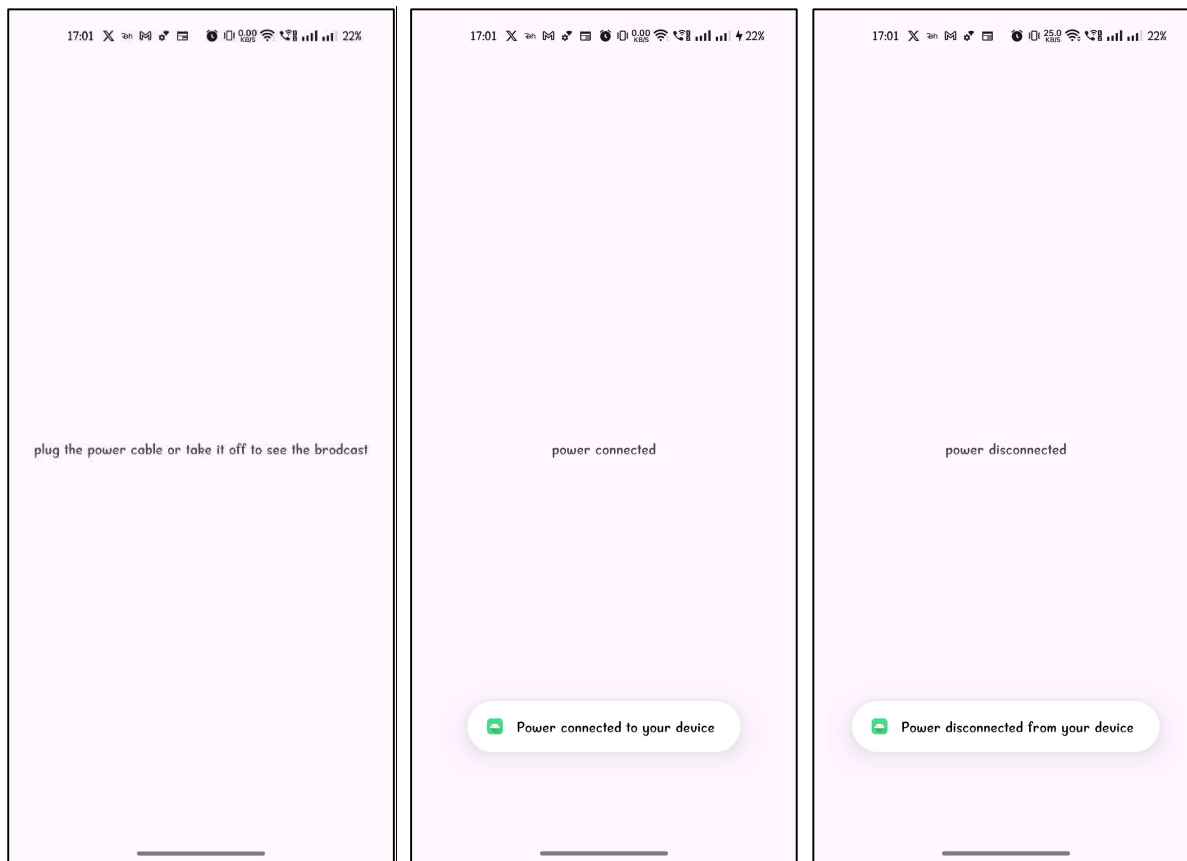
    class Practical12_broadcast extends BroadcastReceiver {
        @Override
        public void onReceive(Context context, Intent intent)
```

```

{
    String action = intent.getAction();

    if (action != null) {
        switch (action) {
            case
"android.intent.action.ACTION_POWER_CONNECTED":
                Toast.makeText(context, "Power connected
to your device", Toast.LENGTH_SHORT).show();
                textView.setText("power connected");
                break;
            case
"android.intent.action.ACTION_POWER_DISCONNECTED
":
                Toast.makeText(context, "Power
disconnected from your device",
Toast.LENGTH_SHORT).show();
                textView.setText("power disconnected");
                break;
            default:
                Toast.makeText(context, "Unknown action:
" + action, Toast.LENGTH_SHORT).show();
                textView.setText("unknown action
triggered");
                break;
        }
    }
}

```



Practical-13

AIM: Develop an Android application that uses Content Providers to share data between different apps and components.

XML:

Manifest file:

Add this to Manifest.xml

```
<provider
    android:name=".Pr13ContentProvider"
    android:authorities="com.mk.madpractical"
    android:enabled="true"
    android:exported="true" />
```

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical13_ConstraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".Practical12">

    <TextView
        android:id="@+id/Practical13_textview"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="we are trying to receive context from the context
provider"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.5" />

    <Button
        android:id="@+id/Practical13_button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toBottomOf="@+id/Practical13_textview"
        app:layout_constraintVertical_bias="0.5" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

Pr13ContentProvider.java:

```
package com.mk.madpractical;

import android.content.ContentProvider;
import android.content.ContentValues;
import android.database.Cursor;
```

```
import android.database.MatrixCursor;
import android.net.Uri;

public class Pr13ContentProvider extends ContentProvider {

    public Pr13ContentProvider() {
    }

    @Override
    public int delete(Uri uri, String selection, String[] selectionArgs) {
        // Implement this to handle requests to delete one or more rows.
        throw new UnsupportedOperationException("Not yet
implemented");
    }

    @Override
    public String getType(Uri uri) {
        // at the given URI.
        throw new UnsupportedOperationException("Not yet
implemented");
    }

    @Override
    public Uri insert(Uri uri, ContentValues values) {
        throw new UnsupportedOperationException("Not yet
implemented");
    }

    @Override
    public boolean onCreate() {
        return false;
    }

    @Override
    public Cursor query(Uri uri, String[] projection, String selection,
String[] selectionArgs, String sortOrder) {
        // Create a Cursor with a single column containing the greeting
message
        MatrixCursor cursor = new MatrixCursor(new
String[]{"message"});
        cursor.addRow(new String[]{"Hello from the MAD Practical 12
Content Provider!"});
        return cursor;
    }

    @Override
    public int update(Uri uri, ContentValues values, String selection,
String[] selectionArgs) {
        throw new UnsupportedOperationException("Not yet
implemented");
    }
}
```

Practical13.java:

```
package com.mk.madpractical;
import android.content.ContentResolver;
import android.database.Cursor;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
```

```

import android.net.Uri;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import androidx.activity.EdgeToEdge;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class Practical13 extends AppCompatActivity {
    Button btn;
    TextView textView;
    private static final Uri CONTENT_URI =
Uri.parse("content://com.mk.madpractical/hello");
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical13);

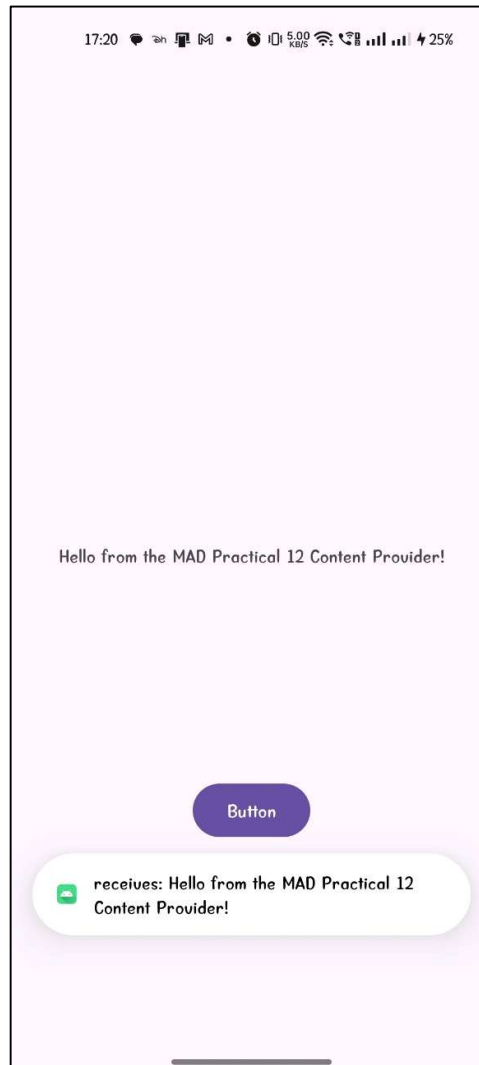
ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.Pr
actical13_ConstraintLayout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,

```

```

systemBars.right, systemBars.bottom);
        return insets;
    });
    btn = findViewById(R.id.Practical13_button);
    textView = findViewById(R.id.Practical13_textview);
    btn.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            ContentResolver contentResolver = getContentResolver();
            Cursor cursor = contentResolver.query(CONTENT_URI, null,
null, null, null);
            if (cursor != null && cursor.moveToFirst()) {
                String msg =
cursor.getString(cursor.getColumnIndex("message"));
                textView.setText(msg);
                Toast.makeText(Practical13.this, "receives: "+msg,
Toast.LENGTH_SHORT).show();
                cursor.close();
            }
        }
    });
}

```



Practical-14

AIM: Develop an Android application that uses Content Providers to read system-level data, such as contacts and calendar events.

XML:

Manifest file:

Add this to Manifest.xml

```
<uses-permission  
    android:name="android.permission.READ_CONTACTS" />
```

Activity.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical14_ConstraintLayout"
android:layout_width="match_parent"
android:layout_height="match_parent"
tools:context=".Practical14">

<ListView
    android:id="@+id/Practical14_listview"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintVertical_bias="0.5" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
o package com.mk.madpractical;
```

```
import android.Manifest;
import android.content.ContentResolver;
import android.content.pm.PackageManager;
import android.database.Cursor;
import android.net.Uri;
import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;


import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;


import java.util.ArrayList;
```

```
public class Practical14 extends AppCompatActivity {

    private static final int PERMISSION_REQUEST_CODE = 100;
```

```

private ListView contactsListView;
private ArrayList<String> contactsList;
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_practical14);

    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.practical14_ConstraintLayout), (v, insets) -> {
        Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
        return insets;
    });

    contactsListView = findViewById(R.id.practical14_listview);
    contactsList = new ArrayList<>();

    if (ContextCompat.checkSelfPermission(this,
Manifest.permission.READ_CONTACTS)
        != PackageManager.PERMISSION_GRANTED) {
        ActivityCompat.requestPermissions(this, new
String[]{Manifest.permission.READ_CONTACTS,
PERMISSION_REQUEST_CODE};
    } else {
        loadContacts();
    }
}

private void loadContacts() {
    ContentResolver contentResolver = getContentResolver();
    Uri uri =
android.provider.ContactsContract.CommonDataKinds.Phone.CONTENT_URI;
    Cursor cursor = contentResolver.query(uri, null, null, null, null);

    if (cursor != null) {
        while (cursor.moveToNext()) {
            String name =
cursor.getString(cursor.getColumnIndex(android.provider.ContactsContract.CommonDataKinds.Phone.DISPLAY_NAME));
            String phoneNumber =
cursor.getString(cursor.getColumnIndex(android.provider.ContactsContract.CommonDataKinds.Phone.NUMBER));
            contactsList.add(name + " " + phoneNumber);
        }
        cursor.close();
    }

    ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
android.R.layout.simple_list_item_1, contactsList);
    contactsListView.setAdapter(adapter);
}

@Override
public void onRequestPermissionsResult(int requestCode,

```

```

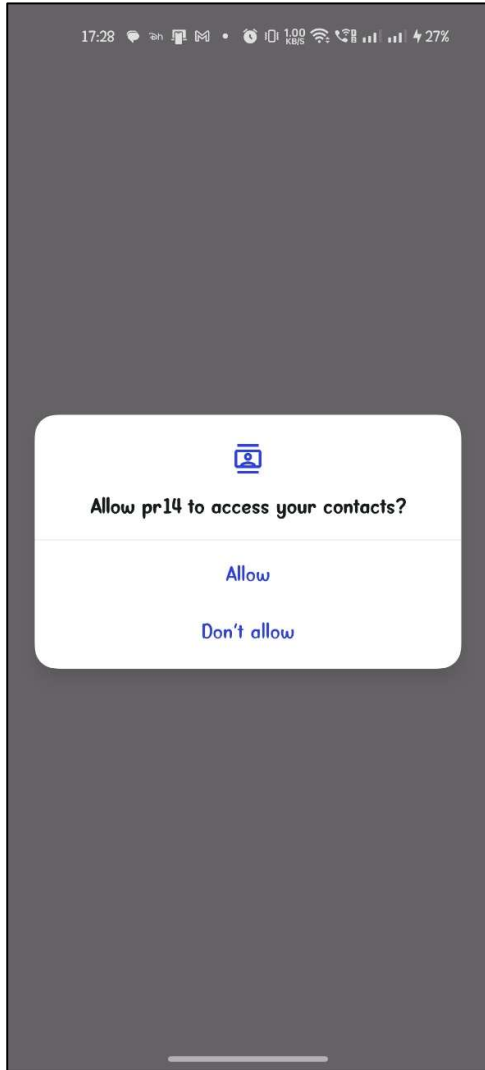
@NonNull String[] permissions, @NonNull int[] grantResults) {
    super.onRequestPermissionsResult(requestCode, permissions,
grantResults);
    if (requestCode == PERMISSION_REQUEST_CODE) {
        if (grantResults.length > 0 && grantResults[0] ==
PackageManager.PERMISSION_GRANTED) {
            loadContacts();

```

```

        } else {
            Toast.makeText(this, "Permission denied to read contacts",
Toast.LENGTH_SHORT).show();
        }
    }
}

```



Practical-15

AIM: Create an application that creates a database using SQLiteOpenHelper Class and performs Insert and Read from the SQLite database.

XML:

View Data activity

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/view_data_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp" >
    <ListView
        android:layout_margin="15dp"
        android:id="@+id/showData"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

Practical15 activity

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical15_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    tools:context=".Practical15">
    <EditText
        android:id="@+id/Pr15RollNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="15dp"
        android:ems="10"
        android:hint="Roll Number"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr15Name"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr15Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="15dp"
        android:ems="10"
```

```
        android:hint="Name"
        android:inputType="text"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr15Semester"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr15Semester"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="15dp"
        android:ems="10"
        android:hint="Semester"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
    />
    <Button
        android:id="@+id/Pr15action_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="15dp"
        android:text="InsertData"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr15Semester"
    />
    <Button
        android:id="@+id/Pr15view_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="View Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr15action_btn"
    />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

DBHelper.java

```
package com.mk.madpractical;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
```

```

import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.widget.AdapterView;
import java.util.ArrayList;

public class DBHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME =
"StudentData.db";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_NAME = "StudentRecord";
    private static final String COLUMN_ROLL = "RollNo";
    private static final String COLUMN_NAME = "name";
    private static final String COLUMN_SEM = "Semester";
    private final Context context;

    public DBHelper(Context context) {
        super(context, DATABASE_NAME, null,
DATABASE_VERSION);
        this.context = context;
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String CREATE_TABLE = "CREATE TABLE IF NOT EXISTS " +
TABLE_NAME + " (" +
            COLUMN_ROLL + " TEXT PRIMARY KEY," +
            COLUMN_NAME + " TEXT," +
            COLUMN_SEM + " TEXT)";
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion,
int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

    public void insertData(String roll, String name, String sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_ROLL, roll);
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);

        db.insertWithOnConflict(TABLE_NAME, null,
contentValues, SQLiteDatabase.CONFLICT_REPLACE);
    }

    public Cursor getDataByRoll(String roll) {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME + "
WHERE " + COLUMN_ROLL + "=?";
        Cursor cursor = db.rawQuery(query, new String[]{roll});
        return cursor;
    }

```

```

    public Cursor getAllData() {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME;
        return db.rawQuery(query, null);
    }

    public boolean updateData(String roll, String name, String
sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);
        int result = db.update(TABLE_NAME, contentValues,
COLUMN_ROLL + "=?", new String[]{roll});
        return result > 0; // Return true if update was successful
    }

    // DELETE operation (Delete data by Roll Number)
    public boolean deleteData(String roll) {
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, COLUMN_ROLL +
"=?", new String[]{roll});
        return result > 0; // Return true if delete was successful
    }

    // DELETE operation (Delete all records)
    public void deleteAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        db.execSQL("DELETE FROM " + TABLE_NAME);
    }

    public ArrayAdapter<String> getAdapter(){
        ArrayList<String> rows = new ArrayList<String>();
        Cursor data = getAllData();
        // data.moveToFirst();
        while(data.moveToNext()){
            String roll =
data.getString(data.getColumnIndex(COLUMN_ROLL));
            String name =
data.getString(data.getColumnIndex(COLUMN_NAME));
            String sem =
data.getString(data.getColumnIndex(COLUMN_SEM));
            String row = "Name: "+name+"\nRoll: "+roll+"\nSem:
"+sem;
            rows.add(row);
        }

        data.close();
        return new ArrayAdapter<String>(context,
androidx.appcompat.R.layout.support_simple_spinner_dropd
own_item, rows);
    }
}

```

ViewData.java

```
package com.mk.madpractical;

import android.os.Bundle;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.AdapterView.OnItemSelectedListener;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;

public class ViewData extends AppCompatActivity {
    private ListView listView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_view_data);
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.view_data_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        listView = findViewById(R.id.showData);
        ArrayAdapter adapter = new DBHelper(this).getAdapter();
        if (adapter.getCount() > 0) {
            listView.setAdapter(adapter);
        } else {
            Toast.makeText(this, "The Database is empty!",
            Toast.LENGTH_SHORT).show();
        }
    }
}
```

MainActivity.java

```
package com.mk.madpractical;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.AdapterView.OnItemSelectedListener;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
```

```
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;
```

```
public class Practical15 extends AppCompatActivity {
    TextView rollNo,name,sem;
    Button action,view;
    ListView listView;
    DBHelper dbHelper;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical15);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical15_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });

        rollNo = findViewById(R.id.Pr15RollNo);
        name = findViewById(R.id.Pr15Name);
        sem = findViewById(R.id.Pr15Semester);
        action = findViewById(R.id.Pr15action_btn);
        view = findViewById(R.id.Pr15view_btn);
        dbHelper = new DBHelper(this);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                dbHelper.insertData(rollNo.getText().toString(),name.getText().
                toString(),sem.getText().toString());
                Toast.makeText(Practical15.this, "Data Inserted
                Sucessfully", Toast.LENGTH_SHORT).show();
                rollNo.setText("");
                name.setText("");
                sem.setText("");
            }
        });

        view.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startActivity(new
                Intent(Practical15.this,ViewData.class));
            }
        });
    }
}
```

Output:

14:00 1 device 2.00 KB 75%

123456

studentname

5

InsertData

View Data

1 2 3 -

4 5 6 ↵

7 8 9 ✕

, 0 . ✓

14:01 1 device 21.0 KB 75%

123456

studentname

5

InsertData

View Data

Data Inserted Sucessfully

14:01 1 device 1.00 KB 75%

Name: moh Roll: 11 Sem: 3

Name: moh Roll: 12 Sem: 3

Name: moh Roll: 13 Sem: 3

Name: moh Roll: 15 Sem: 3

Name: mohit Roll: 16 Sem: 5

Name: moh Roll: 17 Sem: 3

Name: moh Roll: 18 Sem: 3

Name: moh Roll: 19 Sem: 3

Name: studentname Roll: 123456 Sem: 5

Practical-16

AIM: Create an application to Update and Delete data from the SQLite database using SQLiteOpenHelper class

XML:

View Data activity

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

    android:id="@+id/view_data_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp" >
    <ListView
        android:layout_margin="15dp"
        android:id="@+id/showData"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

Practical15 activity

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical16_layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp"
    tools:context=".Practical16">
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="15dp"
        android:text="Enter the Rollno and new name and semester to update the record and enter the Rollno to delete the record"
        app:layout_constraintBottom_toTopOf="@+id/Pr16RollNo"
        tools:layout_editor_absoluteX="10dp" />
    <EditText
        android:id="@+id/Pr16RollNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        android:ems="10"
        android:hint="Roll Number"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr16Name"
        app:layout_constraintEnd_toEndOf="parent"
```

```
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr16Name"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="16dp"
        android:ems="10"
        android:hint="Name"
        android:inputType="text"
        android:text=""
        app:layout_constraintBottom_toTopOf="@+id/Pr16Semester"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="parent" />
    <EditText
        android:id="@+id/Pr16Semester"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:ems="10"
        android:hint="Semester"
        android:inputType="numberDecimal"
        android:text=""
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        />
    <Button
        android:id="@+id/Pr16action_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="16dp"
        android:text="Update Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr16Semester"
        />
    <Button
        android:id="@+id/Pr16delete_btn"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="10dp"
        android:text="Delete Data"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.5"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toBottomOf="@+id/Pr16action_btn"
        />
```

```

<Button
    android:id="@+id/Pr16view_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="10dp"
    android:text="View Data"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/Pr16delete_btn"
"/>
</androidx.constraintlayout.widget.ConstraintLayout>

```

JAVA:

DBHelper.java

```

package com.mk.madpractical;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.widget.AdapterView;
import java.util.ArrayList;
public class DBHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME =
"StudentData.db";
    private static final int DATABASE_VERSION = 1;
    private static final String TABLE_NAME = "StudentRecord";

    private static final String COLUMN_ROLL = "RollNo";
    private static final String COLUMN_NAME = "name";
    private static final String COLUMN_SEM = "Semester";
    private final Context context;

    public DBHelper(Context context) {
        super(context, DATABASE_NAME, null,
DATABASE_VERSION);
        this.context = context;
    }

    @Override
    public void onCreate(SQLiteDatabase db) {
        String CREATE_TABLE = "CREATE TABLE IF NOT EXISTS " +
TABLE_NAME + " (" +
            COLUMN_ROLL + " TEXT PRIMARY KEY," +
            COLUMN_NAME + " TEXT," +
            COLUMN_SEM + " TEXT)";
        db.execSQL(CREATE_TABLE);
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion,
int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }

```

```

    public void insertData(String roll, String name, String sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_ROLL, roll);
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);

        db.insertWithOnConflict(TABLE_NAME, null,
contentValues, SQLiteDatabase.CONFLICT_REPLACE);
    }

    public Cursor getDataByRoll(String roll) {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME + "
WHERE " + COLUMN_ROLL + "=?";
        Cursor cursor = db.rawQuery(query, new String[]{roll});
        return cursor;
    }

    public Cursor getAllData() {
        SQLiteDatabase db = this.getReadableDatabase();
        String query = "SELECT * FROM " + TABLE_NAME;
        return db.rawQuery(query, null);
    }

    public boolean updateData(String roll, String name, String
sem) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put(COLUMN_NAME, name);
        contentValues.put(COLUMN_SEM, sem);

        int result = db.update(TABLE_NAME, contentValues,
COLUMN_ROLL + "=?", new String[]{roll});
        return result > 0; // Return true if update was successful
    }

    // DELETE operation (Delete data by Roll Number)
    public boolean deleteData(String roll) {
        SQLiteDatabase db = this.getWritableDatabase();
        int result = db.delete(TABLE_NAME, COLUMN_ROLL +
"=?", new String[]{roll});
        return result > 0; // Return true if delete was successful
    }

    // DELETE operation (Delete all records)
    public void deleteAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        db.execSQL("DELETE FROM " + TABLE_NAME);
    }

    public ArrayAdapter<String> getAdapter(){
        ArrayList<String> rows = new ArrayList<String>();
        Cursor data = getAllData();
        //    data.moveToFirst();
        while(data.moveToNext()){
            String roll =
data.getString(data.getColumnIndex(COLUMN_ROLL));
            String name =
data.getString(data.getColumnIndex(COLUMN_NAME));
            String sem =
data.getString(data.getColumnIndex(COLUMN_SEM));
            String row = "Name: "+name+"\nRoll: "+roll+"\nSem:

```

```

"+sem;
    rows.add(row);
}
data.close();
return new ArrayAdapter<String>(context,
androidx.appcompat.R.layout.support_simple_spinner_dropdown_item, rows);
}
}

```

ViewData.java

```
package com.mk.madpractical;
```

```

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.util.ArrayList;

```

```

public class ViewData extends AppCompatActivity {
    private ListView listView;

```

```
    @Override
```

```

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_view_data);

```

```

ViewCompat.setOnApplyWindowInsetsListener(findViewById(
R.id.view_data_layout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
    return insets;
});
listView = findViewById(R.id.showData);
ArrayAdapter adapter = new DBHelper(this).getAdapter();
if (adapter.getCount() > 0) {
    listView.setAdapter(adapter);
} else {
    Toast.makeText(this, "The Database is empty!",
Toast.LENGTH_SHORT).show();
}
}
}

```

MainActivity.java

```
package com.mk.madpractical;
```

```

import android.content.Intent;
import android.os.Bundle;

```

```

import android.view.View;
import android.widget.Button;
import android.widget.ListView;
import android.widget.TextView;
import android.widget.Toast;

```

```

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

```

```

public class Practical16 extends AppCompatActivity {
    TextView rollno, name, sem;
    Button action, view, delete;
    DBHelper dbHelper;

```

```
    @Override
```

```

    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical16);

```

```

ViewCompat.setOnApplyWindowInsetsListener(findViewById(
R.id.Practical16_layout), (v, insets) -> {
    Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top,
systemBars.right, systemBars.bottom);
    return insets;
});

```

```

rollno = findViewById(R.id.Pr16RollNo);
name = findViewById(R.id.Pr16Name);
sem = findViewById(R.id.Pr16Semester);
action = findViewById(R.id.Pr16action_btn);
delete = findViewById(R.id.Pr16delete_btn);
view = findViewById(R.id.Pr16view_btn);

```

```
    dbHelper = new DBHelper(this);
```

```

    action.setOnClickListener(new View.OnClickListener() {
        @Override

```

```
        public void onClick(View v) {
```

```

            if (dbHelper.updateData(rollno.getText().toString(),
name.getText().toString(), sem.getText().toString())) {
                Toast.makeText(Practical16.this, "Data Updated
Sucessfully", Toast.LENGTH_SHORT).show();

```

```
            } else {
```

```

                Toast.makeText(Practical16.this, "There is problem
Updating the data may be the RollNo is not found in the
database", Toast.LENGTH_SHORT).show();
            }

```

```
        }
```

```
    });
```

```

    view.setOnClickListener(new View.OnClickListener() {
        @Override

```

```
        public void onClick(View v) {
```

```
            startActivity(new Intent(Practical16.this,
```

```

ViewData.class));
    }
    });
    delete.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            if (dbHelper.deleteData(rollno.getText().toString())){
                Toast.makeText(Practical16.this, "Data deleted
sucessfully", Toast.LENGTH_SHORT).show();
            }else {

```

```

                Toast.makeText(Practical16.this, "There is problem
Deleting the data may be the RollNo is not found in the
database", Toast.LENGTH_SHORT).show();
            }
        }
    });
}
}

```

Output:

The output consists of six screenshots arranged in a 2x3 grid, demonstrating the functionality of an Android application for managing student records.

Top Row:

- Left:** The initial screen with input fields for "Roll Number", "Name", and "Semester". Below the fields are three buttons: "Update Data", "Delete Data", and "View Data".
- Middle:** The "Update Data" button is pressed. The "Name" field now contains "update student".
- Right:** The "Update Data" button is pressed again. A toast message "Data Updated Sucessfully" is displayed at the bottom.

Bottom Row:

- Left:** The "View Data" button is pressed. A list of student records is displayed, including "Name: moh Roll: 11 Sem: 3" through "Name: update student Roll: 123456 Sem: 1".
- Middle:** The "Delete Data" button is pressed. The "Roll Number" field is set to "18".
- Right:** The "Delete Data" button is pressed again. A toast message "Data deleted sucessfully" is displayed at the bottom.

Practical-17

AIM: Perform Firebase Integration to your Android application and store the data in the Firebase Database.

Steps to Perform:

1. Create the project and setup the mainactivity
2. Go to tools -> firebase -> Realtime Database -> Get Started with realtime database java
3. Click on connect it will opens a tab on the browser
4. Now login and create the project (Name of the project will automatically typed)
5. After creating the project click on the connect it will connect the project with firebase project
6. It will create the **google-services.json** in the app directory
7. Now in the firebase console go to build and realtime database
8. Create a realtime database with test mode as security
9. Now the rest of the process is in the code

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"

xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical17_layout"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp"
tools:context=".Practical17">

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter the values that you want to enter in
the Firebase RealTime Database"

app:layout_constraintBottom_toTopOf="@+id/Pr17RollNo"
    android:layout_marginBottom="17dp"
/>

<EditText
    android:id="@+id/Pr17RollNo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="17dp"
    android:ems="10"
    android:hint="Roll Number"
    android:inputType="numberDecimal"
    android:text=""

app:layout_constraintBottom_toTopOf="@+id/Pr17Name"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent" />

<EditText
    android:id="@+id/Pr17Name"
    android:layout_width="match_parent"
```

```
    android:layout_height="wrap_content"
    android:layout_marginBottom="17dp"
    android:ems="10"
    android:hint="Name"
    android:inputType="text"
    android:text=""
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent" />

<EditText
    android:id="@+id/Pr17Semester"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="17dp"
    android:ems="10"
    android:hint="Semester"
    android:inputType="numberDecimal"
    android:text=""
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr17Name"
/>

<Button
    android:id="@+id/Pr17action_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="17dp"
    android:text="InsertData"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr17Semester"
/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA:

```
package com.mk.madpractical;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

import com.google.android.gms.tasks.OnCompleteListener;
import com.google.android.gms.tasks.Task;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;

import java.util.HashMap;

public class Practical17 extends AppCompatActivity {
    TextView rollno, name, sem;
    Button action;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical17);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical17_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        rollno = findViewById(R.id.Pr17RollNo);
        name = findViewById(R.id.Pr17Name);
        sem = findViewById(R.id.Pr17Semester);
        action = findViewById(R.id.Pr17action_btn);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
```

```
public void onClick(View v) {
    String nameval = name.getText().toString();
    String rollnoval = rollno.getText().toString();
    String semval = sem.getText().toString();

    if (rollnoval.isEmpty()) {
        rollno.setError("RollNo Cannot be empty");
        return;
    }
    if (nameval.isEmpty()) {
        name.setError("Name Cannot be empty");
        return;
    }
    if (semval.isEmpty()) {
        sem.setError("Semester Cannot be empty");
        return;
    }
    addData(rollnoval, nameval, semval);
}
});
}

private void addData(String rollnoval, String nameval, String
semval) {
    HashMap<String, String> values = new HashMap<>();
    values.put("rollno", rollnoval);
    values.put("name", nameval);
    values.put("sem", semval);

    FirebaseDatabase database =
    FirebaseDatabase.getInstance();
    DatabaseReference studentsRef =
    database.getReference("Students");
    String key = studentsRef.push().getKey();
    values.put("key", key);

    studentsRef.child(key).setValue(values).addOnCompleteListener
    er(new OnCompleteListener<Void>() {
        @Override
        public void onComplete(@NonNull Task<Void> task) {
            Toast.makeText(Practical17.this, "Data Uploaded
            Sucessfully", Toast.LENGTH_SHORT).show();
            rollno.setText("");
            name.setText("");
            sem.setText("");
        }
    });
}
}
```

Output:

14:30 1 device 100% 78%

Enter the values that you want to enter in the Firebase RealTime Database

Roll Number

Name

Semester

InsertData

14:32 1 device 0.20% 78%

Enter the values that you want to enter in the Firebase RealTime Database

748596

student1

5

InsertData

Data Uploaded Successfully

14:33 1 device 3.00% 78%

Enter the values that you want to enter in the Firebase RealTime Database

748597

student2

5

InsertData

Data Uploaded Successfully

4351604.pdf MAD Practical - Realtime Datab...

console.firebase.google.com/u/0/project/mad-practical-f90f9/database/mad-practical-f90f9-default-rtddb/data/~2F

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Firestore

MAD Practical

Realtime Database Need help with Realtime Database? Ask Gemini

Data Rules Backups Usage Extensions

https://mad-practical-f90f9-default-rtddb.firebaseio.com

Your security rules are defined as public, so anyone can steal, modify or delete data in your database Learn more Dismiss

```

{
  "malav Solanki": {
    "rollno": "226340316063",
    "sem": "5"
  },
  "student1": {
    "rollno": "748596",
    "sem": "5"
  },
  "student2": {
    "rollno": "748597",
    "sem": "5"
  }
}
```

Database location: United States (us-central1)

Practical-18

AIM: Create an application to retrieve data from the Firebase Database and display it in the RecyclerView

Follow the Steps from Practical-17 to Connect with the database

XML:

View layout for RecyclerView:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="10dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18name"
        android:layout_marginLeft="10dp"
        android:textSize="15sp"
        android:text="Name: Student Name"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18rollno"
        android:layout_marginTop="10dp"
        android:layout_marginLeft="10dp"
        android:layout_below="@+id/Pr18name"
        android:text="RollNo: 226340316099"/>
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/Pr18sem"
        android:layout_marginTop="10dp"
        android:layout_marginLeft="10dp"
        android:layout_marginBottom="10dp"
        android:layout_below="@+id/Pr18rollno"
        android:text="Semester: 5"/>
    <View
        android:layout_width="match_parent"
        android:layout_height="1dp"
        android:background="?android:attr/textColorSecondary"
        android:layout_below="@+id/Pr18sem" />
</RelativeLayout>
```

MainActivity:

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/Practical18layout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
tools:context=".Practical18">
```

```
<androidx.recyclerview.widget.RecyclerView
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:id="@+id/Pr18Recyclerview"/>
```

```
</RelativeLayout>
```

JAVA:

Student Class (DataHolder):

```
package com.mk.madpractical;

import java.util.Objects;

public class Student {
    String key; // Firebase key
    String rollno;
    String name;
    String sem;

    public Student(String key, String rollno, String name, String
sem) {
        this.key = key;
        this.rollno = rollno;
        this.name = name;
        this.sem = sem;
    }

    public Student() {
        // Default constructor required for calls to
        dataSnapshot.getValue(Student.class)
    }

    // Getter and setter for key
    public String getKey() {
        return key;
    }

    public void setKey(String key) {
        this.key = key;
    }

    public String getRollno() {
        return rollno;
    }

    public void setRollno(String rollno) {
        this.rollno = rollno;
    }
}
```

```

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public String getSem() {
    return sem;
}

public void setSem(String sem) {
    this.sem = sem;
}
}

```

StudentHolder (ViewHolder for RecyclerView):

```

package com.mk.madpractical;
import android.view.View;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

public class StudentHolder extends RecyclerView.ViewHolder {
    TextView name,rollno,sem;
    public StudentHolder(@NonNull View itemView) {
        super(itemView);
        name = itemView.findViewById(R.id.Pr18name);
        rollno = itemView.findViewById(R.id.Pr18rollno);
        sem = itemView.findViewById(R.id.Pr18sem);
    }
}

```

StudentAdapter (Adapter Class for RecyclerView):

```

package com.mk.madpractical;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.ViewGroup;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;

public class StudentAdapter extends
RecyclerView.Adapter<StudentHolder> {
    public StudentAdapter(Context context, List<Student>
students) {
        this.context = context;
        this.students = students;
    }

    Context context;
    List<Student> students;
    @NonNull

```

```

@Override
    public StudentHolder onCreateViewHolder(@NonNull
ViewGroup parent, int viewType) {
        return new
StudentHolder(LayoutInflater.from(context).inflate(R.layout.pr
18recyclerview,parent,false));
    }

    @Override
    public void onBindViewHolder(@NonNull StudentHolder
holder, int position) {
        Student student = students.get(position);
        holder.rollno.setText("RollNO: "+student.rollno);
        holder.name.setText("Name: "+student.name);
        holder.sem.setText("Semester: "+student.sem);
    }

    @Override
    public int getItemCount() {
        return students.size();
    }

    public void addStudent(Student student) {
        if (getStudentPositionByKey(student.key) == -1) {
            students.add(student);
            notifyItemInserted(students.size() - 1);
        }
    }

    public void updateStudent(Student updatedStudent) {
        int position =
getStudentPositionByKey(updatedStudent.key);
        if (position != -1) {
            students.set(position, updatedStudent);
            notifyItemChanged(position);
        }
    }

    public void removeStudent(String key) {
        int position = getStudentPositionByKey(key);
        if (position != -1) {
            students.remove(position);
            notifyItemRemoved(position);
        }
    }

    private int getStudentPositionByKey(String key) {
        for (int i = 0; i < students.size(); i++) {
            if (students.get(i).getKey().equals(key)) {
                return i;
            }
        }
        return -1;
    }
}

```

Main Activity:

```
package com.mk.madpractical;

import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import com.google.firebase.Firebase;
import com.google.firebase.database.ChildEventListener;
import com.google.firebase.database.DataSnapshot;
import com.google.firebase.database.DatabaseError;
import com.google.firebase.database.DatabaseReference;
import com.google.firebase.database.FirebaseDatabase;
import com.google.firebase.database.ValueEventListener;
import java.util.ArrayList;
import java.util.List;

public class Practical18 extends AppCompatActivity {
    RecyclerView recyclerView;

    private ArrayList<Student> students;
    private StudentAdapter adapter;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical18);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical18layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });
        recyclerView = findViewById(R.id.Pr18Recyclerview);
        recyclerView.setLayoutManager(new
        LinearLayoutManager(this));
        students = new ArrayList<Student>();
        adapter = new StudentAdapter(this, students);
        recyclerView.setAdapter(adapter);
        DatabaseReference databaseReference =
        FirebaseDatabase.getInstance().getReference("Students");

        databaseReference.addChildEventListener(new
```

```
ChildEventListener() {

    @Override
    public void onChildAdded(@NonNull DataSnapshot
    dataSnapshot, String previousChildName) {
        // Called when a new student is added to the
        database
        Student student =
        dataSnapshot.getValue(Student.class);
        if (student != null) {
            student.setKey(dataSnapshot.getKey());
            adapter.addStudent(student);
        }
    }

    @Override
    public void onChildChanged(DataSnapshot
    dataSnapshot, String previousChildName) {
        // Called when a student is updated in the database
        String key = dataSnapshot.getKey();
        Student updatedStudent =
        dataSnapshot.getValue(Student.class);
        if (updatedStudent != null) {
            updatedStudent.setKey(key); // Set the key
            adapter.updateStudent(updatedStudent);
        }
    }

    @Override
    public void onChildRemoved(DataSnapshot
    dataSnapshot) {
        // Called when a student is removed from the
        database
        String key = dataSnapshot.getKey();
        adapter.removeStudent(key);
    }

    @Override
    public void onChildMoved(DataSnapshot dataSnapshot,
    String previousChildName) {
        // Not used in this case
    }

    @Override
    public void onCancelled(DatabaseError databaseError) {
        // Handle possible errors
    }
    });
}
```

Output:

14:59 1 device 0.88 KB/S LTE 4G 79% Name: Mohit Makwana RollNO: 226340316036 Semester: 5 Name: Makwana Mohit RollNO: 226340316036 Semester: 5 Name: Neha Kavaiya RollNO: 226340316029 Semester: 5 Name: Mohit Makwana RollNO: 235 Semester: 6 Name: malav Solanki RollNO: 226340316063 Semester: 5 Name: student1 RollNO: 748596 Semester: 5 Name: student2 RollNO: 748597 Semester: 5	14:59 1 device 5.00 KB/S LTE 4G 79% Name: MONIT MARWANA RollNO: 226340316036 Semester: 5 Name: Makwana Mohit RollNO: 226340316036 Semester: 5 Name: Neha Kavaiya RollNO: 226340316029 Semester: 5 Name: Mohit Makwana RollNO: 235 Semester: 6 Name: malav Solanki RollNO: 226340316063 Semester: 5 Name: student1 RollNO: 748596 Semester: 5 Name: student2 RollNO: 748597 Semester: 5
--	--

Practical-19

AIM: Connect an Android application to the MySQL database using PHP, and insert the data in the database table.

Steps to follow:

1. Start the XAMPP Control Panel
2. Start the apache and mysql server from it
3. In mysql create database user mad@localhost , create database mad , and create table Student with rollno name and sem column
4. Now in xampp/htdocs folder create a php file MAD-Pr19.php and write the code given below
5. Test the php file by typing url in browser with your ip
6. Connect your mobile with same wifi network
7. Test the url in your mobile
8. Finally the app will work

PHP:

```
<?php
$servername = "localhost"; // Database server
$username = "mad"; // Database username
$password = "hellomad"; // Database password
$dbname = "maD"; // Database name

// Create connection
$conn = new mysqli($servername, $username, $password,
$dbname);
if (empty($_POST)){
    die("post method required");
}
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Get POST data from Android app
$rollno = $_POST['rollno'];
$name = $_POST['name'];
$sem = $_POST['sem'];
$sql = "INSERT INTO Student (rollno, name, sem) VALUES
('$rollno', '$name', $sem)";

if ($conn->query($sql) === TRUE) {
    echo "Data inserted successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
?>
```

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical19_layout"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp"
tools:context=".Practical19">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter the values that you want to enter in
the Firebase RealTime Database"
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr19RollNo"
    android:layout_marginBottom="15dp"
/>
```

```
<EditText
    android:id="@+id/Pr19RollNo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Roll Number"
    android:inputType="numberDecimal"
    android:text=""
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr19Name"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent" />
```

```
<EditText
    android:id="@+id/Pr19Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Name"
    android:inputType="text"
    android:text=""
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
```

```

        app:layout_constraintStart_toStartOf="parent" />

<EditText
    android:id="@+id/Pr19Semester"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:ems="10"
    android:hint="Semester"
    android:inputType="numberDecimal"
    android:text=""
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr19Name"
/>

<Button
    android:id="@+id/Pr19action_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="InsertData"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr19Semester"
/>

</androidx.constraintlayout.widget.ConstraintLayout>

```

JAVA:

```

package com.mk.madpractical;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.OutputStream;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.HashMap;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

```

```

public class Practical19 extends AppCompatActivity {
    private ExecutorService executorService;
    private Handler mainHandler;
    TextView rollno, name, sem;
    Button action;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical19);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical19_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });

        executorService = Executors.newSingleThreadExecutor();
        mainHandler = new Handler(Looper.getMainLooper());
        rollno = findViewById(R.id.Pr19RollNo);
        name = findViewById(R.id.Pr19Name);
        sem = findViewById(R.id.Pr19Semester);
        action = findViewById(R.id.Pr19action_btn);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String nameval = name.getText().toString();
                String rollnoval = rollno.getText().toString();
                String semval = sem.getText().toString();

                if (rollnoval.isEmpty()) {
                    rollno.setError("RollNo Cannot be empty");
                    return;
                }
                if (nameval.isEmpty()) {
                    name.setError("Name Cannot be empty");
                    return;
                }
                if (semval.isEmpty()) {
                    sem.setError("Semester Cannot be empty");
                    return;
                }
                insertData(rollnoval, nameval, semval);
            }
        });
    }

    private void insertData(final String rollnoval, final String
    nameval, final String semval) {
        executorService.execute(new Runnable() {
            @Override
            public void run() {

```

```

try {
    // Make the HTTP request in the background thread
    URL url = new
URL("http://192.168.79.127:5700/MAD-Pr19.php");
    HttpURLConnection conn = (HttpURLConnection)
url.openConnection();
    conn.setRequestMethod("POST");
    conn.setDoOutput(true);
    conn.setDoInput(true);
    conn.setRequestProperty("Content-Type",
"application/x-www-form-urlencoded");

    // Prepare POST data
    String postData = "rollno=" + rollnoval + "&name="
+ nameval + "&sem=" + semval;

    // Send POST data
    OutputStream os = conn.getOutputStream();
    os.write(postData.getBytes());
    os.flush();
    os.close();

    // Get the response from the server
    BufferedReader in = new BufferedReader(new
InputStreamReader(conn.getInputStream()));
    String inputLine;
    StringBuilder response = new StringBuilder();

    while ((inputLine = in.readLine()) != null) {
        response.append(inputLine);
    }
    in.close();

```

```

// Send the server response to the main thread
final String result = response.toString();
mainHandler.post(new Runnable() {
    @Override
    public void run() {
        // Show the server response as a Toast on the
main thread
        Toast.makeText(getApplicationContext(),
result, Toast.LENGTH_LONG).show();
        rollno.setText("");
        name.setText("");
        sem.setText("");
    }
});

} catch (Exception e) {
    Log.e("Error", e.getMessage());

    // Post the error message to the main thread
mainHandler.post(new Runnable() {
    @Override
    public void run() {
        Toast.makeText(getApplicationContext(), "Error
occurred:" + e.getMessage(), Toast.LENGTH_LONG).show();
    }
});
}
}
}
}

```

Output:

15:32 1 device 3.00 MB 82%

Enter the values that you want to enter in the Firebase RealTime Database

Roll Number

Name

Semester

InsertData

1 2 3 -

4 5 6 *

7 8 9 ✕

, 0 . ✓

Data inserted successfully

15:34 1 device 2.00 MB 82%

Enter the values that you want to enter in the Firebase RealTime Database

748595

student1

5

InsertData

1 2 3 -

4 5 6 *

7 8 9 ✕

, 0 . ✓

Data inserted successfully

15:34 1 device 1.70 MB 82%

Enter the values that you want to enter in the Firebase RealTime Database

748596

student2

5

InsertData

1 2 3 -

4 5 6 *

7 8 9 ✕

, 0 . ✓

Data inserted successfully

15:34 1 device 2.20 MB 82%

Enter the values that you want to enter in the Firebase RealTime Database

748597

student3

5

InsertData

1 2 3 -

4 5 6 *

7 8 9 ✕

, 0 . ✓

Data inserted successfully

15:34 1 device 2.20 MB 82%

Enter the values that you want to enter in the Firebase RealTime Database

748598

student4

5

InsertData

1 2 3 -

4 5 6 *

7 8 9 ✕

, 0 . ✓

Data inserted successfully

15:38 1 device 3.00 MB 82%

Enter the values that you want to enter in the Firebase RealTime Database

748599

student5

5

InsertData

1 2 3 -

4 5 6 *

7 8 9 ✕

, 0 . ✓

Data inserted successfully

```
mysql>
mysql> select * from Student;
+-----+-----+-----+
| rollno | name   | sem |
+-----+-----+-----+
| 748595 | student1 | 5 |
| 748596 | student2 | 5 |
| 748597 | student3 | 5 |
| 748598 | student4 | 5 |
| 748599 | student5 | 5 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```

Practical-20

AIM: Perform insertion of data to MySQL database using PHP from an Android application.

Steps to follow:

1. Start the XAMPP Control Panel
2. Start the apache and mysql server from it
3. In mysql create database user mad@localhost , create database mad , and create table Student with rollno name and sem column
4. Now in xampp/htdocs folder create a php file MAD-Pr20.php and write the code given below
5. Test the php file by typing url in browser with your ip
6. Connect your mobile with same wifi network
7. Test the url in your mobile
8. Finally the app will work

PHP:

```
<?php
$servername = "localhost"; // Database server
$username = "mad"; // Database username
$password = "hellomad"; // Database password
$dbname = "maD"; // Database name

// Create connection
$conn = new mysqli($servername, $username, $password,
$dbname);
if (empty($_POST)){
    die("post method required");
}
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

// Get POST data from Android app
$rollno = $_POST['rollno'];
$name = $_POST['name'];
$sem = $_POST['sem'];
$sql = "INSERT INTO Student (rollno, name, sem) VALUES
('$rollno', '$name', $sem)";

if ($conn->query($sql) === TRUE) {
    echo "Data inserted successfully";
} else {
    echo "Error: " . $sql . "<br>" . $conn->error;
}
$conn->close();
?>
```

XML:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"

xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:id="@+id/Practical20_layout"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp"
tools:context=".Practical20">
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Enter the values that you want to enter in
the Firebase RealTime Database"
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr20RollNo"
    android:layout_marginBottom="15dp"
/>
```

```
<EditText
    android:id="@+id/Pr20RollNo"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Roll Number"
    android:inputType="numberDecimal"
    android:text=""
```

```
app:layout_constraintBottom_toTopOf="@+id/Pr20Name"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
    app:layout_constraintStart_toStartOf="parent" />
```

```
<EditText
    android:id="@+id/Pr20Name"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginBottom="15dp"
    android:ems="10"
    android:hint="Name"
    android:inputType="text"
    android:text=""
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="1.0"
```

```

        app:layout_constraintStart_toStartOf="parent" />

<EditText
    android:id="@+id/Pr20Semester"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:ems="10"
    android:hint="Semester"
    android:inputType="numberDecimal"
    android:text=""
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr20Name"
/>

<Button
    android:id="@+id/Pr20action_btn"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="15dp"
    android:text="InsertData"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.5"
    app:layout_constraintStart_toStartOf="parent"

app:layout_constraintTop_toBottomOf="@+id/Pr20Semester"
/>

</androidx.constraintlayout.widget.ConstraintLayout>

```

JAVA:

```

package com.mk.madpractical;
import android.os.Bundle;
import android.os.Handler;
import android.os.Looper;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import java.io.BufferedReader;
import java.io.InputStreamReader;
import java.io.OutputStream;
import java.net.HttpURLConnection;
import java.net.URL;
import java.util.HashMap;
import java.util.concurrent.ExecutorService;
import java.util.concurrent.Executors;

```

```

public class Practical19 extends AppCompatActivity {
    private ExecutorService executorService;
    private Handler mainHandler;
    TextView rollno, name, sem;
    Button action;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_practical20);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(
            R.id.Practical19_layout), (v, insets) -> {
            Insets systemBars =
            insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
            systemBars.right, systemBars.bottom);
            return insets;
        });

        executorService = Executors.newSingleThreadExecutor();
        mainHandler = new Handler(Looper.getMainLooper());
        rollno = findViewById(R.id.Pr20RollNo);
        name = findViewById(R.id.Pr20Name);
        sem = findViewById(R.id.Pr20Semester);
        action = findViewById(R.id.Pr20action_btn);
        action.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String nameval = name.getText().toString();
                String rollnoval = rollno.getText().toString();
                String semval = sem.getText().toString();

                if (rollnoval.isEmpty()) {
                    rollno.setError("RollNo Cannot be empty");
                    return;
                }
                if (nameval.isEmpty()) {
                    name.setError("Name Cannot be empty");
                    return;
                }
                if (semval.isEmpty()) {
                    sem.setError("Semester Cannot be empty");
                    return;
                }
                insertData(rollnoval, nameval, semval);
            }
        });
    }

    private void insertData(final String rollnoval, final String
    nameval, final String semval) {
        executorService.execute(new Runnable() {
            @Override
            public void run() {

```

```

try {
    // Make the HTTP request in the background thread
    URL url = new
URL("http://192.168.79.127:5700/MAD-Pr20.php");
    HttpURLConnection conn = (HttpURLConnection)
url.openConnection();
    conn.setRequestMethod("POST");
    conn.setDoOutput(true);
    conn.setDoInput(true);
    conn.setRequestProperty("Content-Type",
"application/x-www-form-urlencoded");

    // Prepare POST data
    String postData = "rollno=" + rollnoval + "&name="
+ nameval + "&sem=" + semval;

    // Send POST data
    OutputStream os = conn.getOutputStream();
    os.write(postData.getBytes());
    os.flush();
    os.close();

    // Get the response from the server
    BufferedReader in = new BufferedReader(new
InputStreamReader(conn.getInputStream()));
    String inputLine;
    StringBuilder response = new StringBuilder();

    while ((inputLine = in.readLine()) != null) {
        response.append(inputLine);
    }
    in.close();

```

```

// Send the server response to the main thread
final String result = response.toString();
mainHandler.post(new Runnable() {
    @Override
    public void run() {
        // Show the server response as a Toast on the
main thread
        Toast.makeText(getApplicationContext(),
result, Toast.LENGTH_LONG).show();
        rollno.setText("");
        name.setText("");
        sem.setText("");
    }
});

} catch (Exception e) {
    Log.e("Error", e.getMessage());

    // Post the error message to the main thread
mainHandler.post(new Runnable() {
    @Override
    public void run() {
        Toast.makeText(getApplicationContext(), "Error
occurred:" + e.getMessage(), Toast.LENGTH_LONG).show();
    }
});
}
}
}
}

```

Output:

Enter the values that you want to enter in the Firebase RealTime Database

Roll Number

Name

Semester

InsertData

1 2 3 -

4 5 6 *

7 8 9 /

, 0 . ✓

Data inserted successfully

15:32 1 device 3.00 88% 82%

15:34 1 device 2.00 88% 82%

15:34 1 device 170 88% 82%

15:34 1 device 2.00 88% 82%

15:38 1 device 3.00 88% 82%

15:38 1 device 3.00 88% 82%

```
mysql>
mysql> select * from Student;
+-----+-----+-----+
| rollno | name   | sem |
+-----+-----+-----+
| 748595 | student1 | 5 |
| 748596 | student2 | 5 |
| 748597 | student3 | 5 |
| 748598 | student4 | 5 |
| 748599 | student5 | 5 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> |
```